

End-only + Mylar-wrapped EJ-204 bar:
Optical simulation study
Geant4 11.4.0 / SSLG4-OPSim

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Motivation & scope

- ▶ **Controlled optical study:** readout *only* at the two ends ($\pm X$ faces); all other faces wrapped in Mylar. Not a final detector configuration.
- ▶ **Goals:**
 - ▶ Isolate end-readout optical response.
 - ▶ Characterize multi-component attenuation with Mylar.
 - ▶ Benchmark timing estimators in a clean geometry.
- ▶ 31 beam positions ($x = -690 \dots +690$ mm), 2,000 μ^- events per position.

This study

End-only + Mylar = extreme limit. Photons may reach an end SiPM, escape the instrumented optical system, or be absorbed in the scintillator bulk or at optical boundaries.

No TB counterpart

This configuration has **no direct test-beam counterpart** (April-2026 TB used top readout + Tyvek).

Electronic jitter not included

SPTR (≈ 106 ps) and FastIC (≈ 10 ps) are post-processing steps, not folded into σ_t shown here.

Geometry & simulation setup

- ▶ **Bar:** EJ-204 (OPSC-101), $1400 \times 60 \times 10 \text{ mm}^3$; long axis X .
- ▶ **End-left SiPMs** (IDs 0–7): $-X$ face.
End-right SiPMs (IDs 8–15): $+X$ face.
- ▶ **Wrapping:** Mylar dielectric surface on $\pm Y, \pm Z$. $R = 0.90$, specular lobe = 1.0, $\sigma_\alpha = 0.1^\circ$.
- ▶ **SUM4:** leading-edge at 4 PE; groups $\{0-3\}/\{4-7\}$ (left), $\{8-11\}/\{12-15\}$ (right).
- ▶ **Rayleigh scattering:** active ($\lambda_{\text{Rayl}} \approx 150 \text{ cm}$ at peak; $\propto \lambda^4$; Materials.cc:45–55).

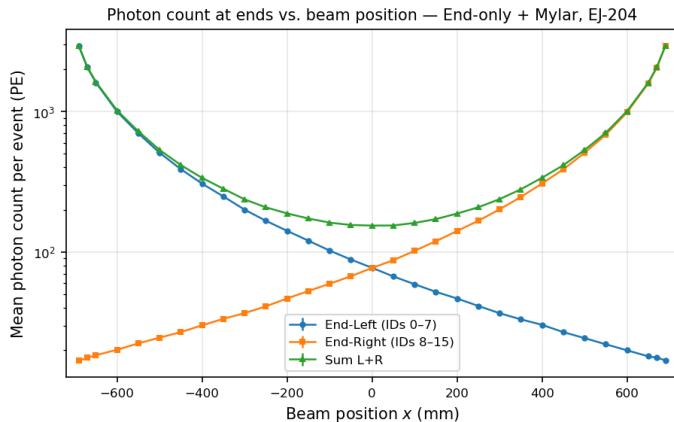
Key parameters

Scint. yield	10,400 γ/MeV
Emission peak	408 nm
Bulk λ (ABSLENGTH)	160 cm
n (flat)	1.58
$v_g = c/n$	189.7 mm/ns
τ_r/τ_d	0.7 ns/1.8 ns
Mylar R	0.90

Note on v_g

No GROUPVEL property set. Geant4 computes $v_g = c/n = 189.7 \text{ mm/ns}$ from flat $n = 1.58$.

Photon yield at ends vs. beam position



Representative values

Near **left** end ($x = -690$ mm):

End-Left (near) ≈ 2926 PE

End-Right (far) ≈ 16.8 PE

Center ($x = 0$):

End-Left ≈ 77.2 PE

End-Right ≈ 77.0 PE

Near **right** end ($x = +690$ mm):

End-Right (near) ≈ 2914 PE

End-Left (far) ≈ 17.0 PE

Two-scale structure

Near-end: prompt contribution.

Center: ≈ 77.2 PE residual
sustained by Mylar recirculation
(many bounces at $R = 0.90$).

Bulk attenuation and longitudinal light-yield scales

ABSLENGTH = 160 cm (material property)

Describes absorption along the **actual optical path length** s : $P_{\text{bulk}}(s) = e^{-s/\lambda_{\text{bulk}}}$. A photon propagating 70 cm longitudinally may traverse significantly more path via reflections.

ABSLENGTH knows nothing about bar geometry or reflectors.

Fitted longitudinal scale (system observable)

NPE vs. distance d combines: bulk absorption (over actual s), Rayleigh scattering, geometry, repeated Mylar surface losses, SiPM acceptance, and PDE. **Any fitted scale is a system-dependent empirical parameter.**

Why single-exp fails

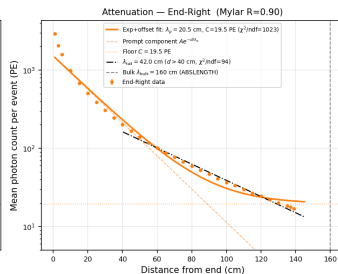
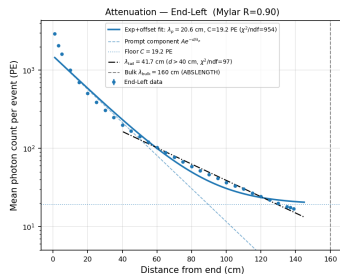
Three numbers inconsistent with $N_0 e^{-d/\lambda}$:

1. Two-point: $\lambda \approx 19$ cm
2. Fit: $N_0 = 811$ PE, $\lambda = 32.2 \pm 1.9$ cm
3. Observed at $d \approx 0$: 2926 PE

Items (2) and (3) disagree by $\approx 3.6\times$.

$\chi^2/\text{ndf} = 2321$: catastrophic. **Consistent with:** a rapidly varying near-end contribution plus a more slowly varying long-distance photon population.

Attenuation: four models and empirical tail slope



M3 result (exp + offset)

$$N(d) = Ae^{-d/\lambda_p} + C$$

$\lambda_p = 20.6 \pm 1.1$ cm (prompt scale)

$C = 19.2 \pm 2.0$ PE (finite-range offset)

$$\chi^2/\text{ndf} = 954$$

M4 (empirical tail slope)

$d > 40$ cm:

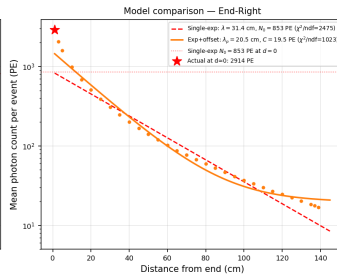
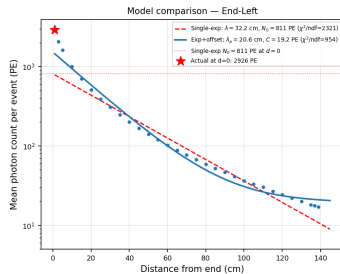
$$\lambda_{\text{tail}} = 41.7 \pm 0.9$$
 cm

$$\chi^2/\text{ndf} = 96.5$$

All $\chi^2/\text{ndf} \gg 1$

All models are **empirical approximations**. λ_p , λ_{tail} are **system-dependent scales**, not the EJ-204 bulk ABSLENGTH (160 cm).

Model comparison: M1, M2, M3, M4 (left side)



Fit diagnostics

Model	λ (cm)	χ^2/ndf
M1: single-exp	32.2 ± 1.9 cm	2321
M3: exp+offset λ_p	20.6 ± 1.1 cm	954
M2: dbl-exp λ_s	9.5 ± 0.4 cm	95
M2: dbl-exp λ_l	43.0 ± 0.9 cm	95
M4: tail ($d > 40$ cm)	41.7 ± 0.9 cm	96.5

Inconsistency

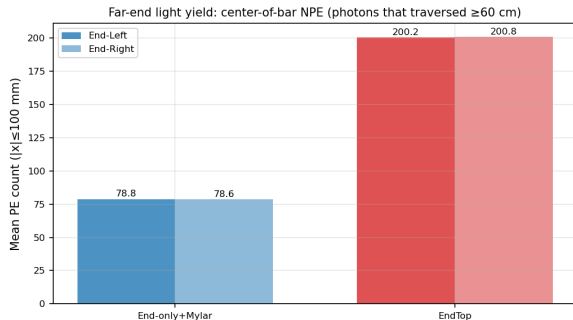
Red star: observed NPE

≈ 2926 PE at $d \approx 0$.

M1 $N_0 = 811$ PE: factor ≈ 3.6 off.

All models have $\chi^2/\text{ndf} \gg 1$:
none is an exact description.

Far-end light yield: where optical boundaries matter



Central light-yield metric (see source CSV definition)

Config	R	NPE metric
End-only+Mylar	0.90	77.2 PE
EndTop	0.98	194.4 PE

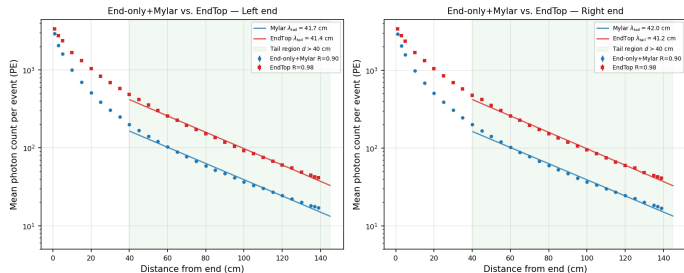
Ratio EndTop/Mylar $\approx 2.5\times$.

Interpretation

Both configs see photons after ≥ 60 cm propagation and many reflections. At Mylar $R = 0.90$: loss per interaction $1 - R = 0.10$ (10%). Cumulative surface losses are expected to become important after many interactions and are consistent with the suppressed long-distance light collection.

The observed ratio is a configuration-level optical-boundary result.

End-only+Mylar vs. EndTop (end SiPMs only)



Identical pipeline

Both datasets: same end-SiPM-only selection, same models, same tail-fit range $d > 40$ cm.

Empirical tail slope ($d > 40$ cm)

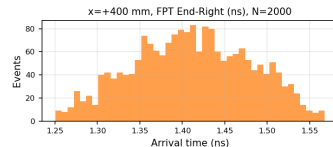
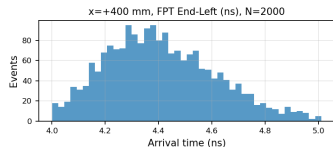
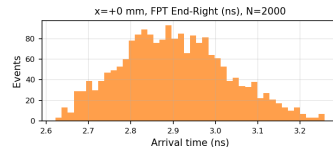
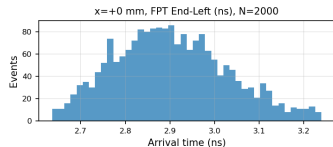
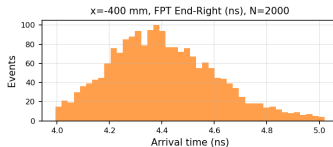
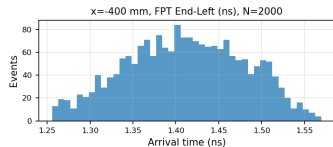
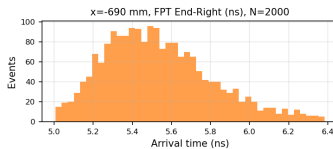
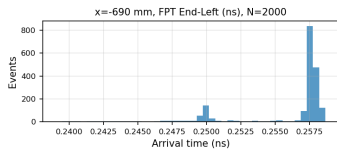
Config	λ_{tail}
End-only+Mylar	41.7 ± 0.9 cm
EndTop (end SiPMs)	41.4 ± 0.7 cm

Numerically similar slopes in this fit range. Does *not* establish reflector independence or exclude model error.

Key conclusion

NPE amplitude at center differs $2.5\times$. Within this selected range, the empirical tail slope provides little discrimination; the absolute end-SiPM yield shows the larger configuration-level difference.

Arrival-time distributions: FPT at representative positions



Distributions shown

First Photon Time (FPT) at
End-Left and End-Right.

Positions: $x \in$

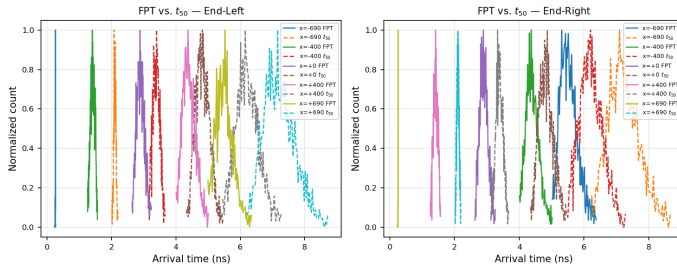
$\{-690, -400, 0, +400, +690\}$ mm.

$N = 2,000$ events/position.

Shape

Distribution shifts to later t and
broadens as beam moves away
from the respective SiPM: longer
path and fewer photons
(statistical broadening of the
minimum).

FPT vs. t_{50} : estimator comparison



FPT vs. t_{50}

FPT: minimum arrival time per event.

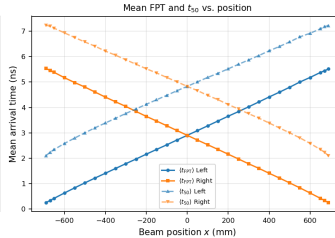
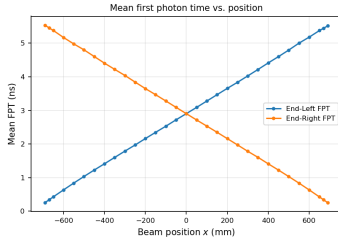
t_{50} : per-event median — less sensitive to photon-population size effects. t_{50} is systematically

later than FPT; offset grows with distance.

Population dependence

FPT and t_{50} sample different parts of a position-dependent arrival-time distribution. Their slopes combine photon-path selection, photon-population size, thresholding, and estimator definition. Neither slope is a direct measurement of photon group velocity.

Position dependence of timing estimators



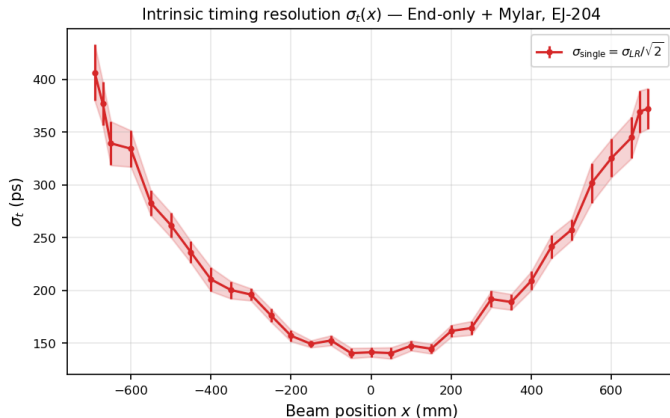
Apparent estimator velocities

Estimator	v_{app}
SUM4 Δt	213.7 mm/ns
FPT slope	264.1 mm/ns
t_{50} slope	275.7 mm/ns
c/n (theory)	189.7 mm/ns

All exceed c/n

These are estimator-dependent apparent responses, not direct measurements of photon group velocity. All exceed $c/n = 189.7 \text{ mm/ns}$ ($n = 1.58$, flat RINDEX). The present inclusive distributions do not isolate a unique mechanism for this difference; candidate effects include photon-path selection, reflections, thresholding, and estimator definition.

Intrinsic timing metric $\sigma_{\text{eq}}(x)$



Key values of σ_{eq}

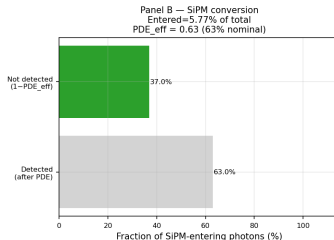
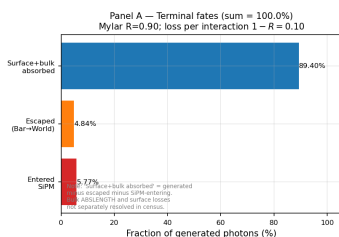
x (mm)	σ_{eq} (ps)
−690	406 ± 27 ps
−400	211 ± 11 ps
0	142 ± 4 ps
+400	209 ± 9 ps
+690	372 ± 19 ps

Definition and caveats

$\sigma_{\text{eq}} \equiv \sigma(t_L - t_R)/\sqrt{2}$ (SUM4, core Gaussian fit). **Equals per-end resolution only under equal, independent-end assumptions** — which break down at the bar ends where NPE is strongly asymmetric. Intrinsic

optical+estimator only. SPTR and electronics **not** included.

Photon budget — End-only + Mylar



Panel A — Terminal fates

Fate	Fraction
Surface+bulk absorbed	$\approx 89.4\%$
Escaped	4.84%
Entered SiPM	5.77%
Total	$\approx 100\%$

Panel B — SiPM conversion

$$\epsilon_{\text{PDE,eff}} = N_{\text{det}} / N_{\text{SiPM}} \approx 0.63$$

Loss note

Mylar loss *per interaction*:

$1 - R = 0.10$. Cumulative surface losses are expected to become important after many interactions. Surface absorption and bulk ABSLENGTH absorption are **not** separately

Summary

Key findings

- ▶ **ABSLENGTH** = 160 cm applies to the actual optical path inside EJ-204. Fitted longitudinal scales ($\lambda_p \approx 20.6 \pm 1.1$ cm, $\lambda_{\text{tail}} \approx 41.7 \pm 0.9$ cm) are system-dependent empirical parameters.
- ▶ **Single-exp fails:** $\chi^2/\text{ndf} = 2321$, N_0 off by $\approx 3.6\times$. M2 is numerically bootstrap-stable but not exact ($\chi^2/\text{ndf} \approx 95$): $\lambda_s = 9.5^{+1.4}_{-0.7}$ cm, $\lambda_l = 43.3^{+2.1}_{-1.6}$ cm.
- ▶ **Far-end NPE** ($d \geq 60$ cm): Mylar ($R = 0.90$): 77.2 PE; EndTop ($R = 0.98$): 194.4 PE ($\approx 2.5\times$ configuration-level difference).
- ▶ **Estimator velocities** all exceed $c/n = 189.7$ mm/ns: estimator-dependent apparent responses, not photon speed.
- ▶ $\sigma_{\text{eq}} = 142 \pm 4$ ps (center), intrinsic only. **Not a 50 ps performance claim.**

Next steps

1. Use $\hat{t}_0 - t_0^{\text{true}}$ as primary metric (direct event-time residual).
2. Apply SPTR *per detected photon* and rerun SUM4; do not add full single-PE SPTR in quadrature.
3. Separate surface from bulk absorption in census.
4. Reflectivity scan ($R = 0.85\text{--}0.95$) in same geometry.
5. Simulate the actual TB geometry before comparing to data.

Appendix A: simulation parameters

Parameter	Value	Unit	Source
Bar material	EJ-204 (OPSC-101)		SSLG4/Materials.cc
Bar dimensions	$1400 \times 60 \times 10$	mm^3	GDML
Readout	End-only ($\pm X$ faces)		geometry
Wrapping ($\pm Y, \pm Z$)	Mylar		run_metadata
Mylar reflectivity R	0.90		run_metadata
Mylar specular lobe	1.0		run_metadata
Mylar σ_α	0.1	deg	run_metadata
Scint. yield	10,400	γ/MeV	Eljen EJ-204 datasheet Aug 2023; Materials.cc:100
Emission peak	408	nm	Eljen EJ-204 datasheet Aug 2023; Materials.cc:86
Bulk λ (ABSLENGTH)	160	cm	Eljen EJ-204 datasheet Aug 2023; Materials.cc:99
Refractive index n	1.58		Eljen EJ-204 datasheet; Materials.cc:41
Density	1.023	g/cm^3	Eljen EJ-204 datasheet; Materials.cc:24
Rise time τ_r	0.7	ns	SSLG4 (EJ-204)
Decay time τ_d	1.8	ns	SSLG4 (EJ-204)
SiPM model (physical)	AFBR-S4N66P014M		hardware
SiPM PDE (peak)	63	% @ 420nm	datasheet
Beam	μ^- , 1 GeV		macro
Positions	31		run_metadata
Events/position	2,000		run_metadata
Geant4 version	11.4.0		log

Appendix B: σ_{eq} per-position summary

x (mm)	N_{trig}	Trig. eff.	σ_{eq} (ps)	Err. (ps)
-690	1911	0.956	406.2	26.6
-400	2000	1.000	210.6	11.2
+0	2000	1.000	141.5	4.3
+400	2000	1.000	209.3	8.9
+690	1911	0.956	372.1	19.1

$\sigma_{\text{eq}} = \sigma(t_L - t_R)/\sqrt{2}$ from SUM4 leading-edge core Gaussian fit. Equals per-end resolution only under equal, independent-end assumptions. Intrinsic optical+estimator only; SPTR and electronics not included.

Appendix C: attenuation fit details (four models)

Model	Side	N_0 or A (PE)	λ (cm)	C (PE)	χ^2/ndf
Single-exp	Left	811 ± 106	32.2 ± 1.9	—	2321
Exp+offset	Left	1502 ± 146	20.6 ± 1.1	19.2 ± 2.0	954
Double-exp short	Left	—	9.5 ± 0.4	—	95
Double-exp long	Left	—	43.0 ± 0.9	—	95
$\lambda_{\text{tail}} (d > 40 \text{ cm})$	Left	429 ± 21	41.7 ± 0.9	—	96.5
Single-exp	Right	853 ± 110	31.4 ± 1.8	—	2475
Exp+offset	Right	1501 ± 142	20.5 ± 1.1	19.5 ± 2.1	1023
$\lambda_{\text{tail}} (d > 40 \text{ cm})$	Right	423 ± 21	42.0 ± 0.9	—	93.6

Weighted fits (per-point SEM). Errors scaled by $\sqrt{\chi^2/\text{ndf}}$ when > 1 . M4 tail: $d > 40 \text{ cm}$. M2 amplitudes are not shown in this compact deck table. All $\chi^2/\text{ndf} \gg 1$: empirical approximations only.

Appendix D: empirical tail-slope range sensitivity

$d_{\min}$	$\lambda_{\text{tail}}^{\text{L}}$	$\lambda_{\text{tail}}^{\text{R}}$	χ^2/ndf (L)
30 cm	39.8 ± 1.0 cm	39.8 ± 1.0 cm	185
40 cm	41.7 ± 0.9 cm	42.0 ± 0.9 cm	97
50 cm	43.5 ± 0.8 cm	43.8 ± 0.8 cm	51

The empirical tail slope increases with $d_{\min}$, indicating residual curvature even in the “tail” region.
Reference: $d_{\min} = 40$ cm.

Appendix E: M2 bootstrap stability (200 replicas, left side)

Parameter	Median	P16	P84
M2: λ_s (cm)	9.5	8.8	10.9
M2: λ_l (cm)	43.3	41.7	45.4
M4: λ_{tail} (cm)	41.8	40.5	43.3

200 bootstrap replicas, seed 42. The resampling unit should be checked in the analysis audit before interpreting these as event-level uncertainties. M2 failure rate: 0%. Numerical stability does not imply model adequacy ($\chi^2/\text{ndf} \approx 95$). λ_l^{M2} is consistent with $\lambda_{\text{tail}}^{\text{M4}}$.

Appendix F: group-velocity and slope-velocity discussion

Theoretical $v_g = c/n$

Flat $n = 1.58$ in `Materials.cc:41`. No explicit GROUPVEL property set (Geant4 computes $v_g = c/n$):

$$v_g = 299.792/1.58 = 189.7 \text{ mm/ns}.$$

Apparent slope velocities

Estimator	v_{app}
SUM4 Δt	213.7 mm/ns
FPT	264.1 mm/ns
t_{50}	275.7 mm/ns
c/n (theory)	189.7 mm/ns

Why these are apparent responses

Slope-derived velocities are estimator-dependent. SUM4 Δt : $v_{\text{app}} = 2/|d\Delta t/dx|$. FPT, t_{50} , and

SUM4 sample different parts of a position-dependent photon ensemble. Their slopes combine photon-path selection, reflections, thresholding, and estimator definition. The present inclusive distributions do not isolate a unique mechanism for $v_{\text{app}} > c/n$. An axial

single-photon injection test (no reflections, known d) is needed to validate Geant4's v_g independently.

Appendix G: method & caveats

- ▶ **Bar geometry:** $1400 \times 60 \times 10 \text{ mm}^3$, axis X . End SiPMs on $\pm X$; Mylar dielectric surface on $\pm Y, \pm Z$.
- ▶ $\sigma_{\text{eq}} = \sigma(t_L - t_R)/\sqrt{2}$ is an *equivalent* per-end metric assuming equal, independent ends. Not a bar-level resolution measurement.
- ▶ **Intrinsic only:** SPTR ($\approx 106 \text{ ps}$) and FastIC ($\approx 10 \text{ ps}$) not folded in. Apply SPTR at the photoelectron level; do not add full SPTR directly in quadrature to event resolution.
- ▶ **SiPM naming:** AFBR-S4N66P024M used only for the PDE curve; physical device is **AFBR-S4N66P014M**.
- ▶ **Photon budget:** surface+bulk absorption not separately resolved in the boundary census; individual Mylar and bulk contributions are therefore not quantified here.
- ▶ **Rayleigh:** active ($\lambda_{\text{Rayl}} \approx 150 \text{ cm}$ at 408 nm , $\propto \lambda^4$); constitutes an additional scattering mechanism beyond ABSLENGTH.
- ▶ **No TB counterpart:** April-2026 TB used top readout + Tyvek $\neq$ end-only + Mylar.
- ▶ **EndTop reference:** EXEC_07, 2,000 events/position, 31 positions. Side reflectors $R = 0.98$ (BarSkinReflector).